MODULE 2: ENERGY STORAGE AND CONVERSION DEVICES (CO: 2)

Complete Comprehensive Textbook: Electrochemical Cells, "Li"-Ion Batteries, Fuel Cells, Solar & DSSC Systems & Solved VIT FAT Question Bank

Aligned with Official VIT BACHY105 Applied Chemistry Course Syllabus, Department Question Bank & VIT FAT Papers

Syllabus Scope & Competency Checklist (Module 2):

- **Electrochemical & Electrolytic Cells:** Fundamental operational differences, thermodynamics ($\Delta G = -nF$ E_cell); electrode nomenclature and conventions; reference electrodes (SHE, SCE, "Ag"/AgCl).
- **The Nernst Equation:** Thermodynamic derivation from reaction quotient Q; concentration cells (electrode and electrolyte); standard cell potential ($E°$); calculations of cell EMF, equilibrium constant (K_eq), and pH.
- **Classification of Modern Batteries:** Primary (irreversible), secondary (rechargeable), and reserve batteries; metrics (energy density, power density, cycle life, self-discharge rate).
- **Lithium-Ion Secondary Batteries:** Electrochemical intercalation/de-intercalation chemistry; cathode materials ($LiCoO_2$, $LiFePO_4$, NMC) and graphitic anode (Li_xC_6); non-aqueous electrolyte ($LiPF_6$ in EC/DMC); Solid Electrolyte Interphase (SEI) formation and passivation; degradation and thermal runaway mechanisms.
- **Fuel Cell Technologies:** Proton Exchange Membrane Fuel Cells (PEMFC) — membrane chemistry (Nafion), electrocatalysis, water management; Solid Oxide Fuel Cells (SOFC) — Yttria-Stabilized Zirconia (YSZ) oxygen ion transport, high-temperature operation, internal reforming.
- **Solar Photovoltaics:** Silicon p-n junction physics; photovoltaic effect; I-V characterization, Short-Circuit Current (I_sc), Open-Circuit Voltage (V_oc), Fill Factor (FF), and Power Conversion Efficiency (η).
- **Photoelectrochemical (PEC) & Dye-Sensitized Solar Cells (DSSC):** PEC solar water splitting at semiconductor photoanodes (TiO_2, Fe_2O_3); Grätzel DSSC structure, ruthenium/organic dye sensitization, electron injection kinetics, iodide/triiodide ($\frac{I^-}{I_3^-}$) redox shuttle, counter electrode electrocatalysis.

1. Electrochemical Cells & The Nernst Equation

1.1 Galvanic vs Electrolytic Cells

An electrochemical cell is a device capable of either generating electrical energy from spontaneous chemical reactions or utilizing electrical energy to cause non-spontaneous chemical transformations:

- **Galvanic (Voltaic) Cells:** Convert chemical energy into electrical energy spontaneously ($\Delta G < 0, \text{E_cell} > 0$).
  - **Anode (Negative terminal):** Site of oxidation (loss of electrons: $M \rightarrow M^{n+} + ne^-$).
  - **Cathode (Positive terminal):** Site of reduction (gain of electrons: $M^{n+} + ne^- \rightarrow M$).
  - **Salt Bridge:** U-tube containing an inert electrolyte (KCl, KNO_3, or NH_4NO_3) in agar-agar gel. Maintains electrical neutrality by counter-ion diffusion and prevents mechanical mixing of the two half-cell solutions while completing the electrical circuit.
- **Electrolytic Cells:** Driven by an external DC power source to force a non-spontaneous reaction ($\Delta G > 0, \text{E_ext} > \text{E_cell}$). Anode is connected to the positive terminal of the power supply; cathode is connected to the negative terminal.
- **Standard Reference Electrodes:**
  1. **Standard Hydrogen Electrode (SHE):** Platinum foil coated with platinum black immersed in 1.0 M H^+ solution with pure H_2 gas bubbled at 1.0 bar. By universal convention, its standard reduction potential is identically zero: $E^\circ\left(\frac{H^+}{H_2}\right) = 0.000$ V at all temperatures.
  2. **Saturated Calomel Electrode (SCE):** Hg | Hg_2 Cl_2 (s) | KCl (sat). Reaction: Hg_2 Cl_2 $(s) + 2e^- \iff 2$ Hg $(l) + 2\text{Cl}^-$. Reduction potential at 25°C is $+0.242$ V vs SHE.
  3. **Silver / Silver Chloride Electrode ("Ag"/AgCl):** Ag | AgCl (s) | KCl (sat). Reduction potential at 25°C is $+0.197$ V vs SHE.

1.2 Derivation of the Nernst Equation

Consider a generalized reversible redox half-cell reaction:

$$aA + bB + ne^- \iff cC + dD$$

From thermodynamics, the Gibbs free energy change under non-standard conditions is related to the standard free energy change ΔG° and the reaction quotient Q:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

where the reaction quotient is $Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$. The electrical work done by an electrochemical cell equals the decrease in Gibbs free energy:

$$\Delta G = -\text{w_elec} = -nF\ \text{E_cell} \quad \text{and} \quad \Delta G^\circ = -nF\ \text{E_cell}^\circ$$

where n is the number of moles of electrons transferred, F is Faraday's constant (96485 C/mol), and E_cell is the electromotive force (EMF). Substituting into the thermodynamic relation:

$$-nF\ \text{E_cell} = -nF\ \text{E_cell}^\circ + RT \ln Q$$

Dividing both sides by $-nF$:

$$\text{E_cell} = \text{E_cell}^\circ - \frac{RT}{nF} \ln Q = \text{E_cell}^\circ - \frac{2.303RT}{nF}\ \text{log_10}\ Q$$

At standard laboratory temperature ($T = 298.15$ K or 25°C):

$$\frac{2.303RT}{F} = \frac{2.303 \times 8.314 \times 298.15}{96485} = 0.05916\ \text{V} \approx 0.0592\ \text{V}$$

The practical engineering form of the Nernst equation is:

$$\text{E_cell} = \text{E_cell}^\circ - \frac{0.0592}{n}\ \text{log_10}\ \left(\frac{[C]^c[D]^d}{[A]^a[B]^b}\right)$$

- **Equilibrium Constant (K__eq):** At dynamic chemical equilibrium, the net cell potential drops to zero (E_cell $= 0$) and the reaction quotient equals the equilibrium constant ($Q =$ K_eq):

$$0 = \text{E_cell}^\circ - \frac{0.0592}{n}\ \text{log_10 K_eq} \quad \Longrightarrow \quad \text{log_10 K_eq} = \frac{n\ \text{E_cell}^\circ}{0.0592}$$

2. Modern Battery Technologies & Lithium-Ion Secondary Batteries

2.1 Battery Classification & Performance Metrics

1. **Primary Batteries (Non-Rechargeable):** Chemical energy is converted irreversibly into electricity. Once reactants are consumed, the battery is dead (e.g., Alkaline $\frac{\text{Zn}}{\text{MnO_2}}$, Zinc-Air, Lithium thionyl chloride).
2. **Secondary Batteries (Rechargeable / Storage Batteries):** Electrochemical reactions are thermodynamically reversible. Discharging transforms active materials; applying an external DC current reverses the chemical transformation, restoring the original active state (e.g., Lead-Acid, "Ni"-Cd, "Ni"-MH, "Li"-ion).
3. **Key Performance Metrics:**
   - **Specific Energy (Energy Density):** Stored energy per unit mass (Wh/kg) or volume (Wh/L).
   - **Specific Power:** Maximum rate of energy discharge per unit mass (W/kg).
   - **Cycle Life:** Number of complete charge-discharge cycles before capacity drops below 80% of rated nominal value.

2.2 Lithium-Ion Battery (LIB) Chemistry & Architecture

Lithium-ion batteries have revolutionized portable electronics and electric vehicles due to their high operating cell voltage ($\approx 3.7 - 4.2$ V), high specific energy ($150 - 280$ Wh/kg), low self-discharge ($< 2\%$ per month), and absence of memory effect.

- **Core Structural Components:**
  1. **Negative Electrode (Anode):** Highly crystalline layered graphite (C_6) coated on a copper foil current collector. Accommodates lithium between graphene sheets through intercalation.
  2. **Positive Electrode (Cathode):** Layered lithium transition metal oxide (LiCoO_2, Li[Ni_x Mn_y Co_z]O_2 NMC, or olivine-structured LiFePO_4) coated on an aluminum foil current collector.
  3. **Electrolyte:** Non-aqueous organic carbonate liquid mixture (Ethylene Carbonate EC + Dimethyl Carbonate DMC) dissolving 1.0 M LiPF_6 (lithium hexafluorophosphate) salt. Aqueous electrolytes cannot be used because lithium reacts violently with water ($E^\circ = -3.04$ V).
  4. **Separator:** Microporous polymeric membrane (polyethylene PE / polypropylene PP, thickness $15 - 25 \mu\text{m}$) that physically isolates anode and cathode to prevent electrical short-circuits while permitting rapid Li^+ ion conduction.

2.3 Electrochemical Intercalation Mechanism

Lithium-ion cells operate as "rocking-chair" batteries where Li^+ ions shuttle back and forth between the interstitial host matrices of the two electrodes:

- **Discharging Mechanism (Spontaneous Power Generation):**
  - **At the Anode (Oxidation):** De-intercalation of lithium from graphite:
$$\mathrm{Li_x\ C_6} \rightarrow 6C + x\mathrm{Li}^+ + xe^-$$
  - **At the Cathode (Reduction):** Intercalation of lithium into transition metal oxide:
$$\mathrm{Li_(1-x)CoO_2} + x\mathrm{Li}^+ + xe^- \rightarrow \mathrm{LiCoO_2}$$
  - **Net Cell Discharge Reaction:**
$$\mathrm{Li_x\ C_6} + \mathrm{Li_(1-x)CoO_2} \rightarrow 6C + \mathrm{LiCoO_2} \quad (\mathrm{E_cell} \approx 3.7\ \mathrm{V})$$
- **Charging Mechanism (External DC Power Applied):** The external charger drives electrons from cathode to anode through the external circuit, forcing Li^+ ions to de-intercalate from the $\mathrm{LiCoO_2}$ lattice, migrate through the liquid electrolyte, and re-intercalate between the graphite layers:
$$\mathrm{LiCoO_2} + 6C \rightarrow \mathrm{Li_(1-x)CoO_2} + \mathrm{Li_x\ C_6}$$

2.4 Solid Electrolyte Interphase (SEI) & Degradation Modes

- **Solid Electrolyte Interphase (SEI) Formation:** During the initial formatting charge cycle, the potential of the graphite anode drops below ≈ 0.8 V vs $\frac{\mathrm{Li}}{\mathrm{Li}^+}$, operating outside the electrochemical stability window of the organic carbonate solvent. The electrolyte decomposes reductively at the graphite surface, forming a passivating nanometer-thin $(10 - 50\ \mathrm{nm})$ composite film comprising inorganic $(\mathrm{Li_2\ CO_3}, \mathrm{LiF}, \mathrm{Li_2\ }O)$ and organic $((\mathrm{CH_2\ OCOO\ Li})_2)$ species.
  - **Protective Role:** The SEI is electrically insulating (prevents continuous solvent reduction and capacity loss) but ionically conductive to Li^+ ions, allowing normal battery operation.
- **Capacity Fade & Degradation Mechanisms:**
  1. **Lithium Plating & Dendrite Growth:** Under fast charging at low temperatures, metallic lithium deposits on the graphite surface, forming needle-like dendrites that can pierce the separator, causing catastrophic internal short circuits and thermal runaway.
  2. **Transition Metal Dissolution:** Trace moisture reacts with $\mathrm{LiPF_6}$ to form hydrofluoric acid (HF), which leaches manganese or cobalt ions from the cathode, destroying the crystalline structure.
  3. **Thermal Runaway:** Exothermic breakdown of the SEI layer above $80 - 120°\mathrm{C}$, followed by reaction of intercalated lithium with electrolyte and oxygen release from the cathode above 180°C, leading to self-sustaining fire and explosion.

3. Fuel Cells: PEMFC & Solid Oxide Fuel Cells (SOFC)

3.1 Principles of Fuel Cell Operation

A fuel cell is an open electrochemical energy converter that continuously transforms the chemical energy of a fuel (hydrogen, natural gas, methanol) and an oxidant (oxygen from air) directly into electrical power without intermediate combustion:

$$\mathrm{Efficiency\ eta_theo} = \frac{\Delta G}{\Delta H} \times 100\%$$

Unlike heat engines governed by Carnot cycle thermal limits ($\eta < 1 - \frac{\mathrm{T_C}}{\mathrm{T_H}}$), theoretical fuel cell efficiencies can exceed $80 - 90\%$ because they are direct isothermal electrochemical converters.

3.2 Proton Exchange Membrane Fuel Cell (PEMFC)

Operating at low temperatures $(60 - 80°\mathrm{C})$, PEMFCs are the preferred power source for automotive hydrogen vehicles (e.g., Toyota Mirai) and portable clean power.

- **Membrane Electrode Assembly (MEA) Structure:**
  - **Electrolyte:** Solid perfluorosulfonic acid polymer membrane (**Nafion**). The polytetrafluoroethylene (PTFE) backbone provides mechanical and chemical robustness; pendant perfluorovinyl ether chains terminated by sulfonic acid groups $(-\mathrm{SO_3}^- H^+)$ form hydrophilic nano-channels. When hydrated with liquid water, these channels facilitate ultra-rapid proton hopping (H^+ conductivity $\approx 0.1\ \mathrm{S/cm}$).
  - **Catalyst Layers:** Platinum nanoparticles $(2 - 4\ \mathrm{nm})$ dispersed on high surface area carbon black $(\frac{\mathrm{Pt}}{C})$.

- **Electrode Reactions:**
  - **Anode (Hydrogen Oxidation Reaction - HOR):**

$$\mathrm{H_2} \rightarrow 2H^+ + 2e^- \quad (E^\circ = 0.000\ \mathrm{V})$$

  - **Cathode (Oxygen Reduction Reaction - ORR):**

$$\frac{1}{2}\ \mathrm{O_2} + 2H^+ + 2e^- \rightarrow \mathrm{H_2}\ O(l) \quad (E^\circ = +1.229\ \mathrm{V})$$

  - **Overall Cell Reaction:**

$$\mathrm{H_2} + \frac{1}{2}\ \mathrm{O_2} \rightarrow \mathrm{H_2}\ O(l) \quad (\mathrm{E_cell}^\circ = 1.229\ \mathrm{V},\quad \Delta G^\circ = -237.1\ \mathrm{kJ/mol})$$

- **Operational Challenges:** Platinum poisoning by carbon monoxide ($\mathrm{CO} > 10$ ppm), and water management (membrane drying causes high ohmic resistance; flooding blocks reactant gas transport).

3.3 Solid Oxide Fuel Cells (SOFC)

Operating at high temperatures ($700 - 1000$°C), SOFCs provide high-efficiency stationary power generation with fuel flexibility.

- **Component Materials:**
  - **Electrolyte:** Ceramic membrane of **Yttria-Stabilized Zirconia (YSZ)**, consisting of $\mathrm{ZrO_2}$ doped with $8 - 10$ mol% $\mathrm{Y_2\ O_3}$. Substituting Y^{3+} for Zr^{4+} creates oxygen vacancies ($\mathrm{V_O}^{\mathrm{bullet\ bullet}}$) in the fluorite crystal lattice, enabling rapid transport of oxide ions (O^{2-}) at high temperatures.
  - **Anode:** Nickel-YSZ cermet ($\mathrm{Ni} - \mathrm{YSZ}$). Nickel acts as electronic conductor and catalytic cracker for hydrocarbon reforming; YSZ inhibits sintering and matches thermal expansion.
  - **Cathode:** Strontium-doped Lanthanum Manganite ($\mathrm{La_}(1-x)\mathrm{Sr_x\ MnO_3}$ LSM).
- **Electrode Reactions:**
  - **Cathode:** $\frac{1}{2}\ \mathrm{O_2} + 2e^- \rightarrow O^{2-}(\mathrm{migrates\ through\ YSZ\ electrolyte})$
  - **Anode:** $\mathrm{H_2} + O^{2-} \rightarrow \mathrm{H_2}\ O(g) + 2e^-$
- **Advantages:** Operates with non-precious nickel catalysts; capable of direct internal reforming of natural gas ($\mathrm{CH_4} + \mathrm{H_2}\ O \rightarrow \mathrm{CO} + 3\ \mathrm{H_2}$); high-quality waste heat enables combined heat and power (CHP) cogeneration reaching total energy efficiencies $> 80\%$.

4. Solar Photovoltaics, Photoelectrochemical & Dye-Sensitized Cells

4.1 Silicon Photovoltaic (PV) Solar Cells

A solar cell directly converts incident sunlight photons into electricity via the photovoltaic effect in a p-n junction:

1. **Photon Absorption & Carrier Generation:** Photons with energy $h\nu \geq \mathrm{E_g}$ (where bandgap $\mathrm{E_g} = 1.12$ eV for silicon, corresponding to $\lambda \leq 1100$ nm) excite electrons from the valence band to the conduction band, generating electron-hole pairs ($e^- - h^+$).
2. **Charge Carrier Separation:** The strong built-in electric field in the p-n depletion region sweeps photogenerated electrons into the n-layer and holes into the p-layer before recombination can occur.
3. **Current Collection:** Electrons flow through an external load to recombine with holes, producing continuous DC electric power.

- **Key Solar Cell Performance Parameters:**
  - **Short-Circuit Current ($\mathbf{I_sc}$):** Maximum current produced when cell terminals are short-circuited ($V = 0$).
  - **Open-Circuit Voltage ($\mathbf{V_oc}$):** Maximum voltage across cell terminals when net current is zero ($I = 0$).
  - **Maximum Power Point ($\mathbf{P_max = V_mp \times I_mp}$):** Operating point on the I-V curve where power output is maximized.
  - **Fill Factor (FF):** Ratio of actual maximum power to the theoretical product of V_oc and I_sc:

$$\mathrm{FF} = \frac{\mathrm{V_mp\ I_mp}}{\mathrm{V_oc\ I_sc}} = \mathrm{P_}\frac{\mathrm{max}}{\mathrm{V_oc\ I_sc}} \quad (0.75 \leq \mathrm{FF} \leq 0.85\ \ \text{for quality cells})$$

  - **Power Conversion Efficiency (η):**

$$\eta = \mathrm{P_}\frac{\mathrm{max}}{\mathrm{P_}}\mathrm{in} \times 100\% = \frac{\mathrm{V_oc\ I_sc\ FF}}{\mathrm{P_in} \times \mathrm{A_cell}} \times 100\%$$

  where standard solar testing irradiance is $\mathrm{P_in} = 1000\ \mathrm{W/m^2}$ (AM 1.5G spectrum at 25°C).

4.2 Dye-Sensitized Solar Cells (DSSC / Grätzel Cell)

Invented by Michael Grätzel (1991), DSSCs mimic biological photosynthesis by decoupling the photon absorption mechanism from the charge carrier transport mechanism.

- **Architecture:**
  1. **Working Electrode (Anode):** Transparent Fluorine-doped Tin Oxide (FTO) conducting glass coated with a mesoporous film of sintered titanium dioxide (TiO_2, anatase, thickness $10 - 15 \mu\text{m}$). The nanoporous network provides an enormous active surface area ($\approx 1000 \times$ geometric area).
  2. **Molecular Sensitizer (Dye):** Monolayer of ruthenium polypyridyl organometallic complex (e.g., N719 dye) chemically chemisorbed via carboxylate anchoring groups onto the TiO_2 nanoparticle surface.
  3. **Redox Electrolyte:** Organic liquid solvent (acetonitrile) containing an iodide/triiodide ($\frac{I^-}{\text{I_3}^-}$) redox couple.
  4. **Counter Electrode (Cathode):** FTO glass coated with a catalytic nanolayer of platinum (Pt).
- **The 5-Step Operational Kinetic Cycle:**
  1. **Photoexcitation:** Incident photon excites the ground-state dye molecule:
  $$\text{Dye} + h\nu \rightarrow \text{Dye}^*$$
  2. **Ultrafast Electron Injection:** The excited dye injects an electron into the conduction band of TiO_2 within femtoseconds ($\tau < 100$ fs):
  $$\text{Dye}^* \rightarrow \text{Dye}^+ + e^-_{\text{CB(TiO_2)}}$$
  3. **Electron Transport:** Injected electrons percolate via diffusive hopping through the interconnected TiO_2 nanoparticle network to the FTO front current collector, driving the external electrical load.
  4. **Dye Regeneration:** The oxidized dye (Dye^+) is rapidly regenerated back to its ground state by electron transfer from iodide ions in the electrolyte:
  $$2\text{Dye}^+ + 3I^- \rightarrow 2\ \text{Dye} + \text{I_3}^-$$
  5. **Cathodic Regeneration:** The triiodide ions (I_3^-) diffuse across the thin electrolyte gap to the platinum counter electrode, where they are catalytically reduced back to iodide by incoming circuit electrons:
  $$\text{I_3}^- + 2e^- \rightarrow 3I^-$$

  The closed catalytic cycle produces net electrical power without any permanent chemical consumption of dye, electrolyte, or semiconductors.

VIT FAT Solved Question Bank: Step-by-Step Numerical Derivations

FAT Problem 1: Nernst Equation EMF, Concentration Cell & Equilibrium Constant

Question (10 Marks): An electrochemical galvanic cell is constructed at 25°C according to the following cell notation:

$$\mathrm{Zn}\ (s)\ |\ \mathrm{Zn^{2+}}(0.02\ \mathrm{M})\ \|\ \mathrm{Cu^{2+}}(0.50\ \mathrm{M})\ |\ \mathrm{Cu}\ (s)$$

Given standard reduction potentials: $E^\circ\left(\frac{\mathrm{Zn^{2+}}}{\mathrm{Zn}}\right) = -0.763$ V and $E^\circ\left(\frac{\mathrm{Cu^{2+}}}{\mathrm{Cu}}\right) = +0.340$ V. (“i”) Write the balanced spontaneous cell reaction and calculate the standard cell potential $\mathrm{E_cell}^\circ$. (“ii”) Apply the Nernst equation to compute the actual EMF (E_cell) of this cell at 25°C. (“iii”) Calculate the standard Gibbs free energy change ΔG° in kilojoules. (“iv”) Determine the thermodynamic equilibrium constant K_eq of the cell reaction at 25°C.

Complete Step-by-Step Solution:

Step 1: Cell Reactions & Standard Potential:

- Anode (Oxidation): $\mathrm{Zn}\ (s) \rightarrow \mathrm{Zn^{2+}(aq)} + 2e^-$
- Cathode (Reduction): $\mathrm{Cu^{2+}(aq)} + 2e^- \rightarrow \mathrm{Cu}\ (s)$
- Net Cell Reaction: $\mathrm{Zn}\ (s) + \mathrm{Cu^{2+}(aq)} \rightarrow \mathrm{Zn^{2+}(aq)} + \mathrm{Cu}\ (s)$ with $n = 2$ electrons transferred.

$$\mathrm{E_cell}^\circ = \mathrm{E_cathode}^\circ - \mathrm{E_anode}^\circ = +0.340 - (-0.763) = +1.103\ \mathrm{V}$$

Step 2: Apply Nernst Equation:

$$\mathrm{E_cell} = \mathrm{E_cell}^\circ - \frac{0.0592}{n}\ \mathrm{log_10}\left(\frac{[\mathrm{Zn^{2+}}]}{[\mathrm{Cu^{2+}}]}\right)$$

$$\frac{[\mathrm{Zn^{2+}}]}{[\mathrm{Cu^{2+}}]} = \frac{0.02}{0.50} = 0.040$$

$$\mathrm{log_10}(0.040) = \mathrm{log_10}(4 \times 10^{-2}) = 0.6021 - 2 = -1.3979$$

$$\mathrm{E_cell} = 1.103 - \left(\frac{0.0592}{2}\right) \times (-1.3979) = 1.103 - (0.0296 \times (-1.3979)) = 1.103 + 0.04138 = +1.1444\ \mathrm{V}$$

Step 3: Standard Free Energy Change ΔG°:

$$\Delta G^\circ = -nF\ \mathrm{E_cell}^\circ = -2 \times 96485\ \mathrm{C/mol} \times 1.103\ \mathrm{V} = -212846\ \mathrm{J/mol} = -212.85\ \mathrm{kJ/mol}$$

Step 4: Equilibrium Constant K__eq:

$$\mathrm{log_10\ K_eq} = \frac{n\ \mathrm{E_cell}^\circ}{0.0592} = \frac{2 \times 1.103}{0.0592} = \frac{2.206}{0.0592} = 37.2635$$

$$\mathrm{K_eq} = 10^{37.2635} = 10^{0.2635} \times 10^{37} = 1.834 \times 10^{37}$$

Boxed Final Answers:

$$\mathrm{E_cell}^\circ = +1.103\ \mathrm{V},\quad \mathrm{E_cell} = +1.144\ \mathrm{V},\quad \Delta G^\circ = -212.85\ \mathrm{kJ/mol},\quad \mathrm{K_eq} = 1.83 \times 10^{37}.$$

FAT Problem 2: Theoretical Capacity of Lithium-Ion Battery Electrodes

Question (10 Marks): A commercial lithium-ion secondary battery utilizes a layered lithium cobalt oxide cathode (LiCoO_2) and a graphitic carbon anode (C_6). Given Atomic Masses: Li $= 6.94$, Co $= 58.93$, $O = 16.00$, $C =$ 12.011 g/mol, and Faraday's constant $F = 96485$ C/mol $= 26801$ mAh/mol. ("i") Calculate the theoretical specific capacity of the LiCoO_2 cathode in mAh/g. ("ii") In commercial cells, only 50% of lithium can be reversibly de-intercalated ($x = 0.5$) before irreversible crystal structural collapse occurs. Find the practical specific capacity. ("iii") Calculate the theoretical specific capacity of the graphitic anode based on the formation of Li C_6 in mAh/g. ("iv") If a cell operates at a nominal voltage of 3.70 V, determine the theoretical specific energy of the LiCoO_2 material.

Complete Step-by-Step Solution:

Step 1: Theoretical Capacity of LiCoO__2 Cathode: Molar mass of LiCoO_2:

$$\text{M_}(\text{LiCoO}_2) = 6.94 + 58.93 + (2 \times 16.00) = 6.94 + 58.93 + 32.00 = 97.87 \text{ g/mol}$$

Theoretical specific capacity Q_theo:

$$\text{Q_}(\text{theo, cathode}) = \frac{nF}{M} = \frac{1 \times 26801 \text{ mAh/mol}}{97.87 \text{ g/mol}} = 273.84 \text{ mAh/g}$$

Step 2: Practical Reversible Capacity of Cathode: De-intercalating more than $x = 0.5$ causes phase transition from hexagonal to monoclinic and oxygen gas evolution:

$$\text{Q_}(\text{practical}) = 0.50 \times \text{Q_}(\text{theo, cathode}) = 0.50 \times 273.84 = 136.92 \text{ mAh/g}$$

(Matches commercial rating of ≈ 140 mAh/g).

Step 3: Theoretical Capacity of Graphite Anode (Li C__6): Six carbon atoms host one lithium atom ($6C +$ $\text{Li}^+ + e^- \rightarrow$ Li C_6): Molar mass of host graphite matrix ($6C$):

$$\text{M_}(6C) = 6 \times 12.011 = 72.066 \text{ g/mol}$$

$$\text{Q_}(\text{theo, anode}) = \frac{1 \times 26801 \text{ mAh/mol}}{72.066 \text{ g/mol}} = 371.90 \text{ mAh/g}$$

(Matches benchmark graphite capacity of 372 mAh/g).

Step 4: Theoretical Specific Energy of Cathode:

$$\text{Specific Energy} = \text{Q_}(\text{practical}) \times \text{V_nominal} = 136.92 \text{ mAh/g} \times 3.70 \text{ V} = 506.6 \text{ mWh/g} = 506.6 \text{ Wh/kg}$$

Boxed Final Answers:

$$\text{Q_}(\text{theo, cathode}) = 273.8 \text{ mAh/g}, \quad \text{Q_}(\text{practical}) = 136.9 \text{ mAh/g}, \quad \text{Q_}(\text{theo, anode}) = 371.9 \text{ mAh/g}, \quad \text{Specific Energy} = 506.6 \text{ Wh/kg}.$$

FAT Problem 3: Solar PV Cell I-V Fill Factor & Efficiency Calculations

Question (10 Marks): A monocrystalline silicon photovoltaic solar cell with active surface area $A = 25\ \text{cm}^2$ is evaluated under standard test conditions (irradiance $\text{P_in} = 1000\ \text{W/m}^2$, AM 1.5G spectrum at 25°C). The measured output characteristics are:

- Open-Circuit Voltage: $\text{V_oc} = 0.620\ \text{V}$
- Short-Circuit Current: $\text{I_sc} = 0.850\ \text{A}$
- Voltage at Maximum Power Point: $\text{V_mp} = 0.510\ \text{V}$
- Current at Maximum Power Point: $\text{I_mp} = 0.780\ \text{A}$

("i") Calculate the maximum electrical power output P_max produced by the solar cell in Watts. ("ii") Compute the Fill Factor (FF) of this solar cell. ("iii") Determine the total incident solar optical power P_light impinging on the cell surface. ("iv") Calculate the Power Conversion Efficiency (η) of the solar cell.

Complete Step-by-Step Solution:

Step 1: Maximum Electrical Power Output $\mathbf{P_max}$:

$$\text{P_max} = \text{V_mp} \times \text{I_mp} = 0.510\ \text{V} \times 0.780\ \text{A} = 0.3978\ \text{W} = 397.8\ \text{mW}$$

Step 2: Calculate Fill Factor (FF):

$$\text{FF} = \text{P_}\frac{\text{max}}{\text{V_oc} \times \text{I_sc}} = \frac{0.3978}{0.620 \times 0.850} = \frac{0.3978}{0.5270} = 0.7548 \quad (75.48\%)$$

Step 3: Total Incident Solar Optical Power $\mathbf{P_light}$: Cell area in square meters:

$$A = 25\ \text{cm}^2 = 25 \times 10^{-4}\ \text{m}^2 = 2.50 \times 10^{-3}\ \text{m}^2$$

$$\text{P_light} = \text{Irradiance} \times \text{Area} = 1000\ \text{W/m}^2 \times (2.50 \times 10^{-3}\ \text{m}^2) = 2.50\ \text{W}$$

Step 4: Power Conversion Efficiency (η):

$$\eta = \text{P_}\frac{\text{max}}{\text{P_}}\text{light} \times 100\% = \frac{0.3978}{2.50} \times 100\% = 15.912\%$$

Boxed Final Answers:

$$\text{P_max} = 0.398\ \text{W}, \quad \text{FF} = 0.755(75.5\%), \quad \text{P_light} = 2.50\ \text{W}, \quad \eta = 15.91\%.$$